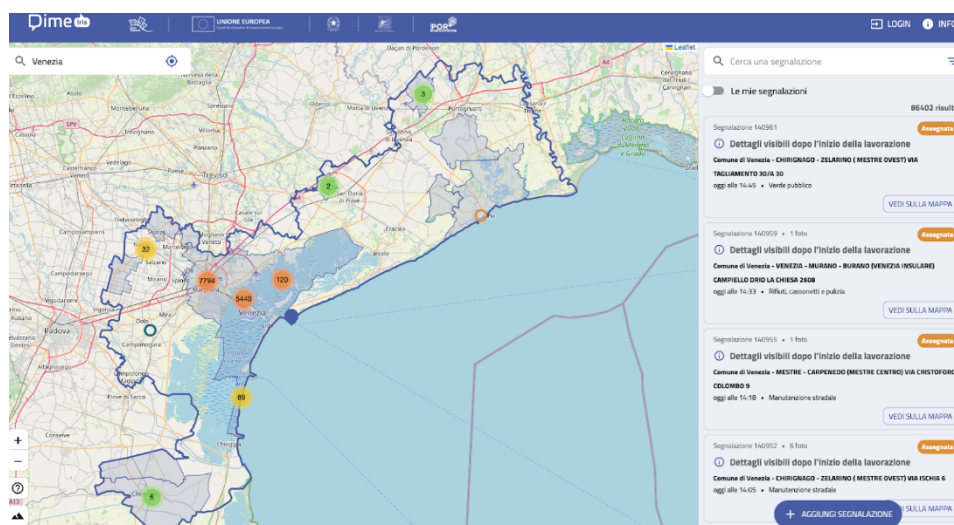


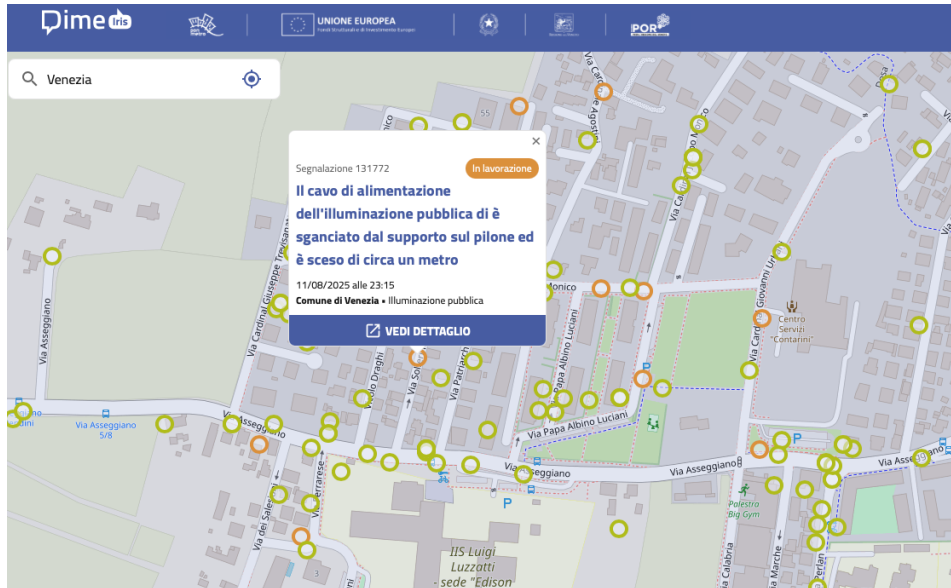
PROJECT – PARTICIPIUM

SYSTEM DESCRIPTION	1
A. REPORTS	2
B. REPORT LIFECYCLE	3
C. CITIZEN UPDATES	3
D. STATISTICS	4
Analysis	5
Delivery Instructions	5

SYSTEM DESCRIPTION

In the Municipality of Turin, a system called **Participium** is being developed — an information system that enables citizen participation in the management of urban environments. It allows citizens to interact with the public administration by reporting inconveniences and malfunctions found in the city (e.g., potholes in the asphalt, sidewalks with architectural barriers, garbage in the streets, broken streetlights, etc.). An example of a similar information system is IRIS in Venice: <https://iris.sad.ve.it/>. The system, once developed, will be made available as open-source software to all Italian public administrations through the portal <https://developers.italia.it/>.





A. REPORTS

Citizens can submit reports only if they have registered in the system with a **username**, **first name**, and **last name**. Once the registration is received, the user gets an email with a confirmation link. The registration becomes valid, and the user can use the system only after confirming through that link. Once registered, a citizen can submit **reports** by selecting a point on the map of Turin (which will be saved with **latitude** and **longitude** values) and filling in a problem form with the following mandatory fields: **title**, **textual description**, **category** (chosen from a predefined list). It is also mandatory to attach one or more photos (up to 3 per report, with each photo stored with its **file path** on the server). The possible **problem categories** are:

Enumeration!!

- Waterworks – Drinking water
- Architectural Barriers
- Sewerage
- Public Lighting
- Waste
- Road Signs and Traffic Lights
- Roads and Urban Furniture
- Public Green Areas and Playgrounds
- Other

After entering all the information and the pictures, the system asks the citizen whether they want the report to be **anonymous** (the name will not appear in the public list of all reports).

B. REPORT LIFECYCLE

Once submitted, the report is in the “Pending Approval” state until the **Organization Office** of the Municipality of Turin performs a **preliminary review** of citizen reports, marking them as **accepted** or rejected. The **possible states** for a report are: **Pending Approval**, **In Progress**, **Work in Progress**, **Suspended**, **Rejected**. After approval, accepted reports move to the “In Progress” state, at which point they are assigned to the **technical office responsible**, based on the problem category. Once the intervention is planned, the state changes to “Work in Progress”, indicating that the issue resolution has started. In some cases, for organizational or technical reasons, the report can be set to “Suspended”, awaiting **further evaluation** or resources. When the problem is solved, the **technical office** updates the status and closes the report. In case of rejection, a written explanation from the Organization Office is mandatory (see next section).

If the intervention must be carried out by maintenance personnel from **an external company** (for example, Enel X for Public Lighting, or company Y for specific reports based on their content), two cases are possible:

- Case 1: The company **has access to Participium**. In this case, the technical office assigns the report to users from the corresponding company. External maintenance personnel can move the report from Assigned to In Progress. Staff from the **technical office** and external maintenance workers **can exchange information and comments through the report**, but these are not visible to the reporting citizen (nor to other citizens). Once the work is completed, the external maintainer can mark the report as **resolved**.

Note: Automatic assignment of all reports in a certain category to the external company (bypassing the initial review by the public relations officer of the municipality) is possible only if previously configured by the municipal administrator.

- Case 2: The company **does NOT have access to Participium**. In this case, the external company updates the technical office outside Participium, and once the problems are solved, a staff member from the technical office will manually move the report to the Resolved state.

C. CITIZEN UPDATES

To strengthen trust between citizens and institutions, the **citizen** can receive **updates** regarding their own reports through various channels. First, at every state change, the citizen receives a **notification** on the platform with the corresponding update. Furthermore, **municipal operators** updating reports can send the reporting citizen a message through the platform, to which it can reply. The system must allow this functionality to be accessible by external chatbots as well. Each time the citizen receives a notification on the platform, they also get an **email** (this option can be disabled in the user settings panel, where they can also upload a personal photo).

Moreover, after the approval phase, accepted reports immediately ^{for both registered and unregistered citizens} become visible on the Participium portal: they appear both on an **interactive map of Turin**, geolocated based on the citizen's selected point, and in a **summary table** that allows filtering and sorting reports by **category**, **status**, or **date**. In both views, the reporter's name ("anonymous" if that option was chosen) and the report title are displayed. Clicking on the title opens a page with the description.

Even non-registered users can view both the maps and the summary table. Once logged into Participium, citizens can follow other citizens' open reports and receive notifications (following the same rules as for their own reports).

D. STATISTICS

The system allows viewing public and private statistics. Public statistics are visible in a dedicated section of the website and concern the number of reports per category, and trends over day, week, or month. They are also visible to non-registered users. In the private section, accessible only to administrators, it is possible to view, in addition to the public statistics, charts and tables on:

- number of reports per state,
- number of reports per type,
- number of reports per type and state,
- number of reports per reporter,
- number of reports per reporter, type, and state,
- number of reports by the top 1% of reporters, per type,
- number of reports by the top 5% of reporters, per type.

Analysis

Analyze the scenario outlined above and define:

1. The conceptual model (UML class diagram) for sections A-B-C.
2. The process model (BPMN process diagram), for sections A-B-C.
3. The use case model (UML use case diagram) at the summary and user-goal level, for all sections.
4. The narrative for the user-goal level use cases identified in the previous point, for sections A-B-C. For section D, write two example narratives for consulting public statistics and two example narratives for consulting private statistics.
5. Evaluate the advantages and disadvantages of developing a web system corresponding to the use cases described in Step 4 by comparing two different alternatives (in both cases consider 250 working days per year, 8 hours of work per day, as a reference).
 1. The development is carried out by a company that provides a team composed of five young people (all under 26 years old) who graduated in computer engineering less than six months ago. They were selected specifically for this project, based on a graduation grade higher than 108 and after a series of tests: a written theoretical test, a programming test in object-oriented languages, and an oral examination. The lead analyst is an experienced person already employed by the company, with over ten years of experience in developing information systems for public administrations. The estimated average cost per team member is 450 euros per day. Apart from the lead analyst, who oversees several projects, all the others work full-time on this project.
 2. The development is also carried out by a company that assigns a team of ten offshore developers plus a lead analyst based in Italy. All individuals involved have considerable experience in similar systems, web programming, and object-oriented languages. Apart from the lead analyst, the developers are located in different parts of Europe in time zones between -1 and +2 hours compared to Turin. The developers and the lead analyst are assigned to the project for one-third of their working time. The average cost per team member is 200 euros per day.

Delivery Instructions

Submission is due by **1:00 p.m. on Friday, January 9, 2026** using the Moodle page related to the project, accessible at <https://moodle.polito.it/mod/assign/view.php?id=11725>.

Premise: Any member of the group whose membership has been entered into the shared sheet can upload files and see files uploaded by other group members. The final delivery can be made by any group member, and the evaluation will be visible to all group members.

Every group will have to upload the following files (exporting them from Signavio where needed):

- **Conceptual model(UML class diagram):**
 - in **PDF** format, called *1-class.pdf*;
 - in **XML** format (see note 1), called *1-class.xml*;
- **Process model (BPMN):**
 - in **PDF** format, called *2-bpmn.pdf*;
 - in **XML** format (see note 1), called *2-bpmn.xml*;
- **Requirements: use case diagram:**
 - in **PDF** format, called *3-ucd.pdf*;
 - in **XML** format (see note 1), called *3-ucd.xml*;
- **Requirements: use case narratives**, in **PDF** format, called *4-ucn.pdf*;
- **Cost estimation (UCP)**, in **Excel** format, called *5-ucp.xlsx* (see note 2);

In addition, it is possible to write in the text box labeled *Online text* comments related **exclusively** to the UCPs. Any comments related to all other diagrams must obligatorily be entered in the respective files (using the annotation functions of Signavio), and will not be considered if left in the text box.

Please note that the absence of any file will result in a null score for the corresponding part of the project.

Notes:

- (1) To export source files to XML from Signavio, refer to the file *XML export guide from Signavio*. The absence of the source file will result in a penalty.
- (2) It is MANDATORY to use the template provided on the Portale della Didattica within the *Project* folder. The different use cases must be added in columns A, B, C, D, E. Use the two separate work tabs for each development alternative. Deviating from the provided template will result in a penalty.